

Rezwan N. Hoque

CONTACT INFORMATION	Brooklyn, NY 11209 (646) 642-7628	rezwannhoque@gmail.com
EDUCATION	Brooklyn College (CUNY) , Brooklyn, NY B.S. in Mathematics (Theoretical Concentration) GPA: 3.980	Expected May 2026
HONORS AND AWARDS	Zimmerberg, Hyman J. '41 Fund for Undergraduate Research in Mathematics (Spring 2024, 2025) Tow Mentoring Initiative Mentee (Spring 2024) Dean's List (since Fall 2022)	
PUBLICATIONS	Sato-Tate Groups and Distributions of $y^\ell = x(x^\ell - 1)$. With Heidi Goodson. Submitted 2024. Preprint: https://arxiv.org/abs/2412.02522	
PRESENTATIONS	Sato-Tate Groups and Distributions of $y^\ell = x(x^\ell - 1)$ AMS-PME Poster Session, Joint Mathematics Meetings (January 2025) Sato-Tate Groups of Aoki Curves Tow Mentoring and Research Student Conference (May 2024) Decimal Expansions: How can we Skip Long Division? CUNY DRP Poster Day (May 2023)	
RESEARCH	Brooklyn College (CUNY) Brooklyn, NY Research Assistant • Collaborated with Prof. Heidi Goodson, PhD • Determined Sato-Tate groups and distributions of nondegenerate curves $y^\ell = x(x^\ell - 1)$ using algebraic and arithmetic geometry • Computed theoretical moment statistics of normalized L-function coefficients via Sage and Python, optimized for $\sim 97\%$ reduction in runtime • Currently investigating degenerate, hyperelliptic curves: $y^2 = x^{p^2} - 1$	January 2024 - Present
SERVICE	Member of Curriculum Committee , Brooklyn College Department of Mathematics (March 2024 - Present) Secretary of Mathematics Club , Brooklyn College (January 2024 - May 2025)	
EMPLOYMENT AND EXPERIENCE	SBS Summer Institute REU Mentee • Worked under Dr. Terrence R. Blackman • Edited upcoming book presenting six African American mathematicians and their technical works CUNY Directed Reading Program Mentee • Studied elementary and algebraic number theory	Brooklyn, NY July 2025 - August 2025 New York, NY January 2023 - May 2023

- Presented findings on retrieving blocks of non-terminating decimal expansions and necessary and sufficient conditions for terminating decimal expansions

Mathnasium

Brooklyn, NY

Tutor

December 2021 - June 2022

- Mentored K-12 students in math via personalized plans